- We were able to come up with over 10 questions for our stakeholder meeting on 9/10.
- We assigned responsible and supporting roles for DH01
- We assigned a designated area for where our files will be found (Teams).
- We worked on finding interviewees for each member.
- We decided on day and time for weekly team meetings for the entire semester.
- We created the 60second project pitch for 9/10 class.

Meet the Stakeholders

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Dr. Robert Read, Dr. Ashley Taylor, Dr. Robert Stone
- Date and Time: 9/10/2026 3:30pm-4:30pm
- Format: Zoom
- Individual notes:
  - We need to contact the Scottland team and them a lot of questions about the machine and what they are doing to get rid of anesthetic waste-gas.
- Team Discussion (**team leader** only):
  - We reassigned an area for where all our files will be found after talking to our stakeholder (GitHub)
  - We were able to get clear expectations for the project and further understand what needs to be completed.
  - We decided on biweekly meetings with our stakeholder and to increase this to weekly in the spring semester.
  - We were able to discuss initial ideas for the project with our stakeholder.
  - We established our future meeting formats, and our stakeholder is very easy to reach by email.

Advisor Meeting

- Attendees: Brooke Struyk, Dr. Robert Stone
- Date and Time: 9/14 9:00am-9:30am
- Format: In-person
- Individual notes:
  - Dr. Stone has a pediatric anesthesiologist if needed for contact
  - Dr. Muelenaer is a pediatrician that can also be contacted (Rachel used him as her interviewee)
  - Dr. Stone reached out for a quote for an IR sensor for any anesthetic gas (sensor looks at specific peaks once asked the input of the gas)
  - Dr. Stone does not want us reading from slides and is okay with 3x5 notecards for presentations
- Team Discussion (**team leader** only):
  - Discussed findings with the team on 9/16
  - Texted Rachel later that day to make sure she got an interview with her stakeholder as early as possible

Weekly Meeting

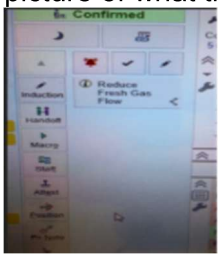
- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/16 6:30pm-8:30pm
- Format: In-person
- Individual notes:
  - Stayed longer at the meeting to talk with Sarah about how we want to split up the design and COMSOL work
  - Finish DH01 over the weekend, so team can start reviewing

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- Team Discussion (**team leader** only):
    - We used this time to work on our DH01 and make sure everyone understood the sections that were required of each other.
    - We also discussed a timeline for our solution and came up with a loose idea of what we all wanted to create.

Meeting with Interviewee

- Attendees: Brooke Struyk, Brian Struyk
- Date and Time: 9/30 6:30pm-7:30pm
- Format: Zoom
- Individual notes:
    - At CHOP and globally, anesthesiologists are talking about reducing the environmental effects of AWG
    - Low-flow is the recommended way the anesthesiologist can reduce the about of AWG entering the air
    - When there is high-flow after a period of 15 minutes, the anesthesia machine will send a message on the notification bar. This bar is checked regularly throughout surgery. A picture of what this looks like is below.



    - There has also been a push to use sevoflurane over Nitrous oxide; however, in low resource countries nitrous oxide is the most common anesthetic because it is cheap.
    - Anesthesiologists typically ensure prior to the surgery that the WAGD system has been turned on and it stays on the entire day.
    - Anesthesiology technicians are the main maintenance team for leaks or issues. If the issue is bigger, BMET would be brought in.
    - The final say on a new technology goes more towards the BMET, and maintenance department is they can account for the new machine.



    - The picture above shows the WAGD device that is being used at CHOP. The white box is a CO2 scrubber, and the flapper valve on the right has a max and min dial to show if there is enough suction for the AWG.
- Team Discussion (**team leader** only):
    - During weekly team meeting on 9/23, I was able to talk about what I learned from my stakeholder.
    - I was also able to provide insight on how pediatric anesthesiologists interact with AWG systems.

Weekly Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/23 6:30pm-7:30pm

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- Format: In-person
- Individual notes:
  - Complete SRR Slide by Friday
  - Prepare for meeting tomorrow with KidsOR and write down any extra questions I may come up with
- Team Discussion (**team leader** only):
  - The team created a list of questions to ask Courtney from KidsOR
  - The team created a meeting time for the SRR video recording over Zoom (decided on Friday)
    - We also made an extra meeting time after the SRR recording due date changed to Sunday. If we needed to re-record, we would be on Sunday at noon (we did not end up needing this meeting).

Meet with KidsOR Team

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Courtney Ludick (KidsOR), Andy Muelenaer, Dr. Robert Stone
- Date and Time: 9/24 3:30pm-4:30pm
- Format: Zoom
- Individual notes:
  - Fix SRR slides based on the meeting
    - Fix Job Map
  - Start SolidWorks tutorials because it looks like there will be more of a need
  - Read over manuals and floor plans sent to us within the next week
  - Read over full page of meeting notes written by Sarah on the Teams site
- Team Discussion (**team leader** only):
  - We were able to get most of our questions answered by Courtney
  - Dr. Muelenaer also asked interesting questions that were helpful for clarity
  - The team realized we had a misunderstanding of what our solution was and what KidsOR actually does.
    - The meeting was very helpful in clearing out confusion after DH01 research
  - The team was able to start coming up with solutions and brainstorming what direction we could go in

SRR Video Recording Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/25 5:30pm-6:30pm
- Format: Teams
- Individual notes:
  - Download the video from teams
  - Edit the video and ensure it does not need to be re-recorded.
  - Submit the final video
- Team Discussion (**team leader** only):
  - We recorded in the video and prepared our slides beforehand, so the meeting was straightforward
  - We discussed future directions with our solution and what Courtney had talked about in the meeting from the day prior
  - We also discussed what we changed in our slides based on the new deliverables

Weekly Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/30 6:30pm-7:30pm

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- Format: Zoom
- Individual notes:
  - Complete the ethics section of DH01
  - Continue to work on SolidWorks tutorials
  - Do extra research on active scavenging systems
- Team Discussion (**team leader** only):
  - Split up DH02
  - Signed the document for the DH02
  - Discussed further what our solution could be and how to prepare for our meeting with Dr. Stone on 10/1.

Weekly Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 10/7 6:30pm-9:30pm
- Format: In-Person
- Individual notes:
  - Start to think of a solution
  - Create a meeting where we discuss individual goals
- Team Discussion (**team leader** only):
  - Read through all the manuals and started to understand where our system could connect
  - This meeting was very eye-opening because we all started to realize where our solution could go and how we could make it and not take up space in the OR
  - We all came into this meeting on ideas of solutions and discussed together what direction we think made the most sense
  - Rachel brought up a condensate idea, but we finalized at the end of the meeting at the usage of adsorption was the best option to get rid of the gas

Weekly Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah
- Date and Time: 10/14 6:30pm-8:00pm
- Format: In-Person
- Individual notes:
  - Prepare for 10/15 meeting
  - Write more questions for stakeholders down on Miro Board
- Team Discussion (**team leader** only):
  - We created and sent questions to our stakeholders prior to the meeting, so they would have the preparation to respond
  - We also further discussed using adsorption as a solution
  - We once again read through the manuals and looked at spare parts and part specs to make sure our solution would be able to connect to the anesthetic machine

Stakeholder Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Courtney Ludick (KidsOR), Dr. Andy Muelenaer, Dr. Robert Read
- Date and Time: 10/15 3:30pm-5:00pm
- Format: Zoom
- Individual notes:
  - Need to look into where the solution we make can go
  - Research different local tube materials that we could use for our system

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    - Create a meeting with Dr. Stone because the stakeholder was convinced our solution could be putting a hole in the wall and putting the tube through it
- Team Discussion (**team leader** only):
    - We were able to ask all the questions we prepared for the team
    - We found that the real problem is that gas isn't moving through the original pipes that have been put on the anesthetic machine, so we need to include a fan system to create more flow
        - We also need to find the best material to use for tubing and can use the comparison of tubing as our COMSOL project
    - We also found out that there is an issue with the pressure relief valve opening
    - We were able to get pictures of the original tubing system that isn't working and found out that in most cases because of the disruption in the room, the tube has been taken out entirely
        - This means anesthetic waste gas has been filling the room freely

Advisor Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Dr. Andy Muelenaer, Dr. Robert Stone
- Date and Time: 10/19 9:00am-10:00am
- Format: Zoom
- Individual notes:
    - Look into different adsorption methods brought up by Dr. Stone
    - Look into ways to add an electrical component
        - Display screen, pH sensor, etc
    - Figure out the diameter and length of the tube we need based on the 4L/min flow requirement
- Team Discussion (**team leader** only):
    - We discussed our issues with what the stakeholder wanted us to complete
    - We also determined together that adsorption would be the best method for a "scavenging system" and that we should continue to make our project a standalone scavenging system
    - Dr. Stone was brought up using wet ping pong balls, kitty litter, and activated charcoal
    - We also determined we needed to reach out to Etta Hanlon, a medical school student who could potentially get us access to anesthetic gases
    - We decided on using either ceramic or PVC piping for the gas
    - We determined we need to start meeting every week with our advisor because he can't make any of the stakeholder meetings and there is a miscommunication issue

Weekly Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 10/21 6:30pm-8:00pm
- Format: In-Person
- Individual notes:
    - Prepare to present the solution to stakeholders
    - Come up with potential clarifying questions to make sure our solution is on the right track
- Team Discussion (**team leader** only):
    - We came up with one singular solution to present to our stakeholders
    - We drew the solution and took a picture
    - Our main goal for this meeting was to make sure our stakeholders and our team were all on the same page about a solution

Stakeholder Meeting

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- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Courtney Ludick (KidsOR), Dr. Andy Muelenaer, Dr. Robert Read
- Date and Time: 10/22 3:30pm-4:30pm
- Format: Zoom
- Individual notes:
  - Find articles proving that adsorption is a viable option to filter the gas
  - Look into check valves
  - Look into backflow because it is important to keep the child safe
  - Moisture from breathing may be an issue, find a way to filter this potential water
- Team Discussion (**team leader** only):
  - The room is cooler than the breath of a child, so water will condense in the tube creating droplets
  - The fan can be plugged into a plug in the back of the anesthetic machine that is 220 V
  - We want to make the design modular in case adsorption doesn't work, we can still use the fan to carry the gas outside or further away
  - We need to look into comparing kitty litter versus activated charcoal to use as an adsorber
    - Could also use for modeling in COMSOL
  - We should add a check valve that the end to prevent backflow
  - Adsorption increases resistance which can be modeled on COMSOL

Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 10/23 5:30pm-6:30pm
- Format: In-Person
- Individual notes:
  - Come up with research questions for Etta Hanlon
  - Complete PDR slides
- Team Discussion (**team leader** only):
  - Emailed Etta Hanlon about meeting and talking about her research using anesthetic gases and looking at the environment
  - We discussed our stakeholder meeting and how we need to change our solution based on the review
  - We split up PDR slides

Prototype Build Meeting

- Attendees: Brooke Struyk, Rachel Mensah
- Date and Time: 10/25 6:15pm-8:45pm
- Format: In-Person
- Individual notes:
  - Share the picture with the team
  - Update the PDR slides
- Team Discussion (**team leader** only):
  - Built a low fidelity prototype (seen in figure 9)
    - Used for the PDR and for help with the CAD modeling

Literature Review

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<It is expected that each team member will read and summarize **three new pieces of literature with each new submission!** Keep track of the total number below by incrementing the header. You can choose from peer-reviewed, scientific articles (https://scholar.google.com/), patents (https://patents.google.com/), engineering standards (request from Dr. Arena ahead of time if not accessible through VT or ANSI), or FDA guidance documents (https://www.fda.gov/regulatory-information/search-fda-guidance-documents#guidancesearch). **Do NOT use literature that was separately assigned for homework.**

Note: AI should serve as a collaborator, not a replacement, for your own critical thought and skills. The intellectual contribution and design decisions must remain your own. AI tools may lack direct access to Google Scholar/Patents and the most recent literature. Additionally, AI-generated content can be inaccurate, biased, or incomplete. Therefore, it is up to you to identify the key pieces of literature for your project. AI can be used to support the Summary of Key Findings section, but you should verify the accuracy of the content and that nothing from a piece of literature was overlooked. Please cite the AI tool used, the prompt that you provided, and your analysis of, or updates to, the content. The other sections should be completed without the use of AI.>

1. Anesthesia and its environmental impact: approaches to minimize exposure to anesthetic gases and reduce waste [1]

Source Authors, Journal, Date, Affiliation

[1] K. Samad, Muhammad Saad Yousuf, H. Ullah, Syed Shabbir Ahmed, Khalid Maudood Siddiqui, and A. Latif, “Anesthesia and Its Environmental Impact: Approaches to Minimize Exposure to Anesthetic Gases and Reduce Waste,” *Medical Gas Research*, vol. 15, no. 1, Jun. 2024, doi: 10.4103/mgr.medgasres-d-23-00059.

Summary of Key Findings

Anesthesia is a mix of nitrous oxides and halogenated ethers that are strong greenhouse gases and contribute to global warming. Operating rooms account for 20-30% of waste leaving the hospital. By improving anesthetic waste gas sensors and collecting systems or changing the flow rate of anesthetic gas, there would be less output. It is important to consider the environmental effect of anesthesia, its contribution to greenhouse gas emissions and global warming.

Application to Design Project

In our initial stakeholder meeting we talked about creating a system that would flow anesthetic gas out of a window. Our stakeholder was unsure what this would mean for the environment if this ended up as a solution, so I decided to use one of my articles to read more about it. The article talks about low-flow anesthesia, so it would help reduce the number of halogenated ethers entering the environment. It would be a good question to ask KidsOR if they are using low-flow anesthesia methods. As of right now the team also uses a chemical bed that collects this fluorine that’s bad for the air and saturates it by forming mineral salts, which might be better for the environment already. This is something we can do further research on.

Specific Information for the Design project

- Acute care settings account for 8% of greenhouse gas emissions

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- United States healthcare sector contributes to nearly 10% of the total GHG emissions with hospitals being the primary source
- Operating rooms generate 20-30% of the entire waste output of a hospital
- Recommeded ways to minimize anesthetic gas waste exposure: LFA, advancing scavenging system, vaporizer, closed-loop delivery system, routine maintenance and calibration, and waste gas monitoring

2. An Initial Evaluation of a Novel Anesthetic Scavenging Interface [2]

Source Authors, Journal, Date, Affiliation

[2] J. A. Barwise, L. J. Lancaster, D. Michaels, J. E. Pope, and J. M. Berry, "An Initial Evaluation of a Novel Anesthetic Scavenging Interface," *Anesthesia & Analgesia*, vol. 113, no. 5, pp. 1064–1067, Nov. 2011, doi: 10.1213/ane.0b013e31822c9a2c.

Summary of Key Findings

The primary method used in the United States and Europe is an active gas scavenging system (AGSS). The machine requires a 24-hour per day period which leads to higher energy costs. By using a DGSS, you do not need a pump as big of a pump, energy costs are lower, and the device can be turned on and off. The article does note that there is no sensor display or failure modes that have been identified.

Application to Design Project

It was brought up during the stakeholder meeting that we should investigate anesthetic scavenging devices as a potential solution, so I decided to read more about it. The article depicts a solution from Vanderbilt University in 2011. If we were to use/ create our own scavenging system this design approach would be a good place to start. The researchers were able to save $28,000 after a 6-month testing period. Though it would also be $2738 per year on electricity. We need to figure out what KidsOR is using and how much it is costing them per year, so we could potentially use a similar solution and optimize it. The device built does not have any display or failure mode, so it's not necessarily what we are doing unless we wanted to start from scratch. This idea might also end up costing KidsOR more money than building onto their existing solution of the bed chamber.

Specific Information for the Design project

- Waste anesthetic gas scavenging technology has not changed much in the past 30 years
- Process requires a large vacuum pump that is expensive and operates continuously
- Standards for waste anesthetic gas were recommended but never implemented
- AGSS works by: "Exhausted gas from the anesthesia machine is removed by an open manifold connected to a vacuum source. The system entrains and evacuates waste gas with relief from positive and negative pressure in the breathing circuit. The balance is achieved either with relief valves that open to atmosphere with a slight pressure gradient or valveless 'tube-within-a-tube' system"
- Many hospitals and ambulatory centers used a dedicated scavenging system that has a separate pump and piping.

3. Waste anesthetic gas exposure and strategies for solution [3]

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Source Authors, Journal, Date, Affiliation

[3] H.-B. Deng, F.-X. Li, Y.-H. Cai, and S.-Y. Xu, “Waste Anesthetic Gas Exposure and Strategies for Solution,” Journal of Anesthesia, vol. 32, no. 2, pp. 269–282, Feb. 2018, doi: 10.1007/s00540-018-2448-1.

Summary of Key Findings

The article brought up that different countries have different standards for how to treat anesthetic waste gas. It also brought up that long term exposure can lead to chronic diseases that based on their research harm the anesthesiologist later on. This is especially concerning because there is no standard biomarker test for elongated waste gas exposure. The article recommends using a scavenging system,, air-conditioning system, Iso-Gard Mask, or passive scavenging system with activated charcoal canister to effectively get rid of gas waste.

Application to Design Project

This article brough up a few interesting points that there is no set limit for the amount of sevoflurane given to a patient, and there is also no universal standard for waste anesthetic gas control. KidsOR travels to different countries and had to abide by different standards, so I think this would be interesting to ask if there is some initial research they have to do before going to a new country and administering anesthesia. The article also brought up concerning biomarkers for WAG exposure after a long period of time. I think it would be interesting to ask the team if they recognize they do any yearly testing for these biomarkers. It would also be interesting to ask my stakeholder, who is a pediatric anesthesiologist, if the hospital he works at has yearly tests.

Specific Information for the Design project

- It’s impossible to fully eliminate waste anesthetic gas (WAG) from the air
- Long term exposure to WAG can lead to increased oxidative stress DNA damage, genotoxicity, and carcinogenesis
    - Some have found an increase in chronic diseases, spontaneous abortion, congenital malformations, low birth rates, and higher stillbirth in offspring
- Check out the National Institute of Occupational Health and Safety and American Conference of Governmental Industrial Hygienists for future literature reviews on WAG exposure
- 28% of OR’s complied with predetermined standards about WAG
- Different countries have their own limit standards (ADD ABOVE)
    - There is a table on page 3 that would be useful to reference exposure limits in different countries
- There is no set limit for the amount of sevoflurane used
- WAG is higher when patient is in a sitting position compared to standing
- Anesthesiologist is the one getting the most amount of WAG exposure compared to the rest of the OR
- There is no “good” biomarker testing WAG exposure levels

4. A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery [4]

Source Authors, Journal, Date, Affiliation

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[4] J. C. Nesbitt, D. A. Krageschmidt, and M. C. Blanco, “A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery,” *Journal of Occupational and Environmental Hygiene*, vol. 10, no. 9, pp. D125-D131, Sep. 2013, doi: 10.1080/15459624.2013.

Summary of Key Findings

When giving laboratory animals, specifically rats, anesthetic gases, in this case isoflurane, researchers noticed excessive concentrations of the gas throughout the surgical site. When using the passive scavenging system, there five times the regular amount of waste gas in the air. The researchers came up with a nose cylinder and an exhausted induction chamber as their solutions. The solutions ended up significantly reducing the amount of isoflurane and oxygen levels to safe levels.

Application to Design Project

One of our initial deliverables was to look at how AWG is treated in veterinary settings. This article brought up two interesting solutions. Instead of making a new scavenging system, they went to the root cause and improved the mask to specifically fit rat’s noses. By also creating a cylinder around the rat’s full head as well, there was no significant/ unsafe amount of gas allowed to escape. The second solution was interesting because it placed the rat in a box and closed the animal in. Once inside, anesthesia was used and scavenged, and then the box was opened. Both solutions are interesting because it looks at the root cause of the issue, that is AWG and solves it. In real-life operating rooms neither of these solutions could work, but it does highlight the importance of an active scavenging system that filters the AWG from the air.

Specific Information for the Design project

- On the chamber the minimum vacuum flow was 17 liters per minute
    - Vacuum evacuated oxygen and isoflurane
- The article brings up excessive amounts of oxygen leading to issues from the AWG, so I will investigate this when we are deciding what specific part of the AWG we want to target
- Potentially further look into anesthesia masks
    - Realistically nothing will be changed about this because it has to be a universal device for all children and sometimes adults who are different sizes, but it is important to see why the AWG escapes
- Some scavenging systems use exhaust rates of 12-15 liters/minute

5. Uptake and biotransformation of sevoflurane in humans: A comparative study of sevoflurane with halothane, enflurane, and isoflurane [5]

Source Authors, Journal, Date, Affiliation

[5] Y. Shiraishi and K. Ikeda, “Uptake and Biotransformation of Sevoflurane in Humans: A Comparative Study of Sevoflurane With Halothane, Enflurane, and Isoflurane,” *Journal of Clinical Anesthesia*, vol. 2, no. 6, pp. 381–386, Nov. 1990, doi: 10.1016/0952-8180(90)90024-W.

Summary of Key Findings

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The article compared 4 different anesthetic gases: sevoflurane, halothane, enflurane, and isoflurane. They tested these gases for an hour on 32 patients who were healthy and monitored the amount that was absorbed by the body, release of fluoride into the blood, and the potential damages on the kidney and liver. They found that sevoflurane raised fluoride levels in the blood the most and had low absorption rates. They found that isoflurane and enflurane was broken down by the body the most. There were also no effects on the kidney or liver by any of the four gases.

Application to Design Project

Before designing our scavenging system, Dr. Stone told us it would be easiest to choose one anesthetic waste gas to be collected by the system. This way we can collect clear data on a specific gas and test for reduction in the air. The team is leaning towards sevoflurane gases because it was brought up by Courtney that this gas runs out the quickest in the hospitals. By targeting this gas, there is a potential to make a big difference in the amount of AWG left in the air. I read this article because I thought it would be important to see the differences in the anesthetic gases being used. It was also interesting to see how the body reacts to different gases used in anesthesia.

Specific Information for the Design project

- If a low amount of sevoflurane is being absorbed by the body, there is a potential that a lot would be breathed back out and contributing to the anesthetic waste gas issue
- It was brought up that the gas used up the quickest is the sevoflurane. This may be due to the low rate of absorption within the body which causes doctors to use excess amounts/ more of it.
    - It would be interesting to further look into why sevoflurane is a popular to use as anesthetic gas even with its low absorption levels.
    - Also look into the other 3 gases and how often they are used in hospital settings
- Sevoflurane required using a higher concentration when comparing it to the other gases, so they would all measure up to 1.1 minimum alveolar concentration (MAC) (the standard the researchers used).

6. Scavenging, Medical Gas Pipelines, and Vacuum Systems [6]

Source Authors, Journal, Date, Affiliation

[6] N. A. Shallik, W. A. Eldegowa, M. O. Zekri, and M. F. Alhammad, “Scavenging, Medical Gas Pipelines, and Vacuum Systems,” *Improving Anesthesia Technical Staff’s Skills*, pp. 137–150, 2022, doi: 10.1007/978-3-030-88849-7_10.

Summary of Key Findings

The article walked one through each part of a waste gas scavenging system. There are four main parts to a scavenging system: collecting tubes, transferring tubes, a scavenging interface, and disposal tubing. In collecting tubes there is an adjustable pressure limiting valve (APL) and a pressure relief valve that collects the extra anesthetic gases. The transferring tubes have a scavenging outlet at the back of the anesthesia machine. The scavenging interface protects the breathing circuit from any positive or negative pressure changes. Finally, the disposal tubing carries the gas from the interface to the disposal. This is the main difference between active and passive systems.

Application to Design Project

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Our main deliverable in the project is to create a standalone active scavenging system, so I thought it would be interesting to read more about the parts of these systems and how it works mechanically. The article walks one through how passive versus active scavenging systems work. It also highlights the importance of active systems and why they are more popularized in hospital settings. It also brings up a potential maintenance schedule for the scavenging machines that we could use and compare to the actual schedule KidsOR has created. The article also brings up why active systems are centrally used throughout the hospital, so by understanding this, we can create a standalone system.

Specific Information for the Design project

- The article recommends the scavenging system of the breathing system should be checked daily.
    - Documented maintenance schedules for the ventilation system in the OR, post-anesthesia care units, and the anesthesia machines
    - Depending on what Courtney sends we could additional information about this, especially for the ventilation PMs that are being used in places with passive systems
- The article recommends the scavenging system of the breathing system should be checked daily.
- Due to the fact medical gases are used in all areas of the hospital, there is a central gases control panel.
- Our solution needs to have a warning and alarm system to ensure that there is no issue in the breathing circuit and AWG levels are not getting too high.

7. Analysis and Failure Prevention of Nozzle Check Valves Used for Protection of Rotating Equipment Due to Wear and Team in the Oil and Gas Industry [7]

Source Authors, Journal, Date, Affiliation

[7] K. Sotoodeh, “Analysis and Failure Prevention of Nozzle Check Valves Used for Protection of Rotating Equipment Due to Wear and Tear in the Oil and Gas Industry,” *Journal of Failure Analysis and Prevention*, vol. 21, no. 4, pp. 1231–1239, May 2021, doi: 10.1007/s11668-021-01162-2.

Summary of Key Findings

The main function of a check valve is to protect rotating mechanical equipment in piping systems from fluid reversal and stopping backflow. Check valves open to add fluid to flow in the intended direction, though adding them to the system can reduce pressure and add to flow resistance. When fluid is moving backwards, the valve will shut, preventing backflow. When gas flow is too little, the control valve may struggle to stay open and cause chattering. This wears and damages the internal system.

Application to Design Project

One of the questions our stakeholders had was about whether our solution takes backflow into account. If moisture or gas gets pushed back into the system, it could get into a child’s lungs and cause harm or death. This issue is super important and with potential moisture issues, the team also needs to account for water. We plan to implement a check valve at the end of our system to make sure issues like backflow are solved. We also need to make sure if we do implement a check valve that is in the correct area (end, middle, beginning), and that enough flow from the fan would have been made by that area for the valve to open.

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Specific Information for the Design project

- Further look into nozzle check valves which is recommended for gas by the article
- Look at different check valve opening flow rates and determine how we can implement it
- When thinking about moisture, maybe we could use a check valve higher up to make sure backflow is not occurring with the gas or moisture
- Look into how a check valve and pressure relief valve can work together and how it impacts backflow and flow in general.

8. A new device to reduce the consumption of halogenated anesthetic agent

Source Authors, Journal, Date, Affiliation

[8] M. Enlund, L. Wiklund, and H. Lambert, “A New Device to Reduce the Consumption of a Halogenated Anaesthetic Agent*,” *Anaesthesia*, vol. 56, no. 5, pp. 429–432, May 2001, doi: 10.1046/j.1365-2044.2001.01900.x.

Summary of Key Findings

This article describes a heat and moisture exchanger that reduces the amount of anesthetic gas needed during surgery. This device was used on 16 patients, and they were specifically using isoflurane. When comparing regular anesthetic gas intake systems to this device there was a 40% reduction in the amount of anesthetic waste gas used. They are collected data on the amount of anesthetic waste gas that enterers into the environment. The reduction while using this device was 55%.

Application to Design Project

There wasn’t much that this article adds to the current solution we have thought about. It is interesting that even though their main goal wasn’t to reduce anesthetic gas waste, they were still able to. This is an interesting solution because long term this solution could be safer for the patient. It could also be used in hospitals that can’t afford to use a lot of anesthetic gases and buy cheap nitrous oxide which is very bad for the environment. It could also help low-resource areas overall save the amount of gases they are using and help prevent anesthetic gas buildup in the air specifically in ORs using passive systems.

Specific Information for the Design project

- They measured the gas before and after the system to test how much was being used
    - We could use this same system for testing when we have created the solution and have gotten access to the anesthetic gases

9. An observational laboratory study to evaluate an anesthetic gas adsorber without anesthetic gas scavenging system

Source Authors, Journal, Date, Affiliation

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[9] K. Nickel *et al.*, "An Observational Laboratory Study to Evaluate an Anesthetic Gas Adsorber Without Anesthetic Gas Scavenging System," *BMC anesthesiology*, vol. 25, no. 1, p. 377, 2025, doi: 10.1186/s12871-025-03223-7.

Summary of Key Findings

This article is looking at if activated carbon can capture sevoflurane without an active scavenging system. The system they are using has activated carbon as its adsorber and CONTRAfluran canisters. The researchers were able to make activated using coconut shells. They noted that the sevoflurane was properly adsorbed by the activated carbon and this system is effective for anesthetic waste gas capturing.

Application to Design Project

The article uses CONTRAfluran that is made from polypropylene and coconut-derived fibers. The anesthetic gas, specifically sevoflurane and desflurane, is captured in the pores of the container. It can also be connected to SENSOfluran that has an LED light and alarm system. When using the researcher's system, it took them 9.6 hours to achieve a 4 L/min flow rate. This is the flow rate we are looking to get with help from the fan system. They also determined that moisture and carbon dioxide made little difference in how the system worked. This was something we were concerned about, but this article is proving that it isn't an issue. There were also warning lights when the canister got too full. The team was looking for a specific alarm or display system, so we could do something like this. We could also look at the area of a specific hospital and use a local activated carbon adsorber for our solution.

Specific Information for the Design project

- Look into different types of activated carbon adsorbers
    - Look at the local area specifically in Malawi and compare pricing and accessibility to kitty litter
- Look into CONTRAfluran canisters and see how our solution could compare
- The canisters use an audible alarm system and lights to signify a change, which is what we could use for our systen

Individual Design Updates

<This section is your electronic design notebook. Include sketches, photos, engineering drawings, equations, calculations, tables, graphs, observations, etc**. Focus on your narration of the design process. For example, your discussion of trainings that you complete and the outputs and/or challenges of that training process (e.g., COMSOL tutorial results, CAD training results, etc.).** Be sure to describe each item in detail with explanatory text so that it is easy to understand when reviewing. It is up to you on how to effectively organize the section headers. Include any notes about updates to User Needs (UN), Design Inputs (DI), Design Outputs (DO), Design Verification (VER), and Design Validation (VAL)**. There should be at least 3 new pages with each new submission**. In select cases, you may paste and circle **individual items** from your Mural or Miro team boards and describe your contribution in the text below**.** However, most of the content in this section should be your **individual work.** Make sure that the font size is large enough to be legible, including within figures**.** However, **a majority of the 3 pages (>=1.5 pages) should be your own descriptive text or narration of the design process. Additionally, it is expected that a majority of the content (>= 1/2) is new or expands beyond what was completed as part of other assignments, such as**

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reading reflections and DHF's. If you include items from a prior individual assignment, please note that in bold/red and expand your discussion.>

Defining User Needs and Purpose

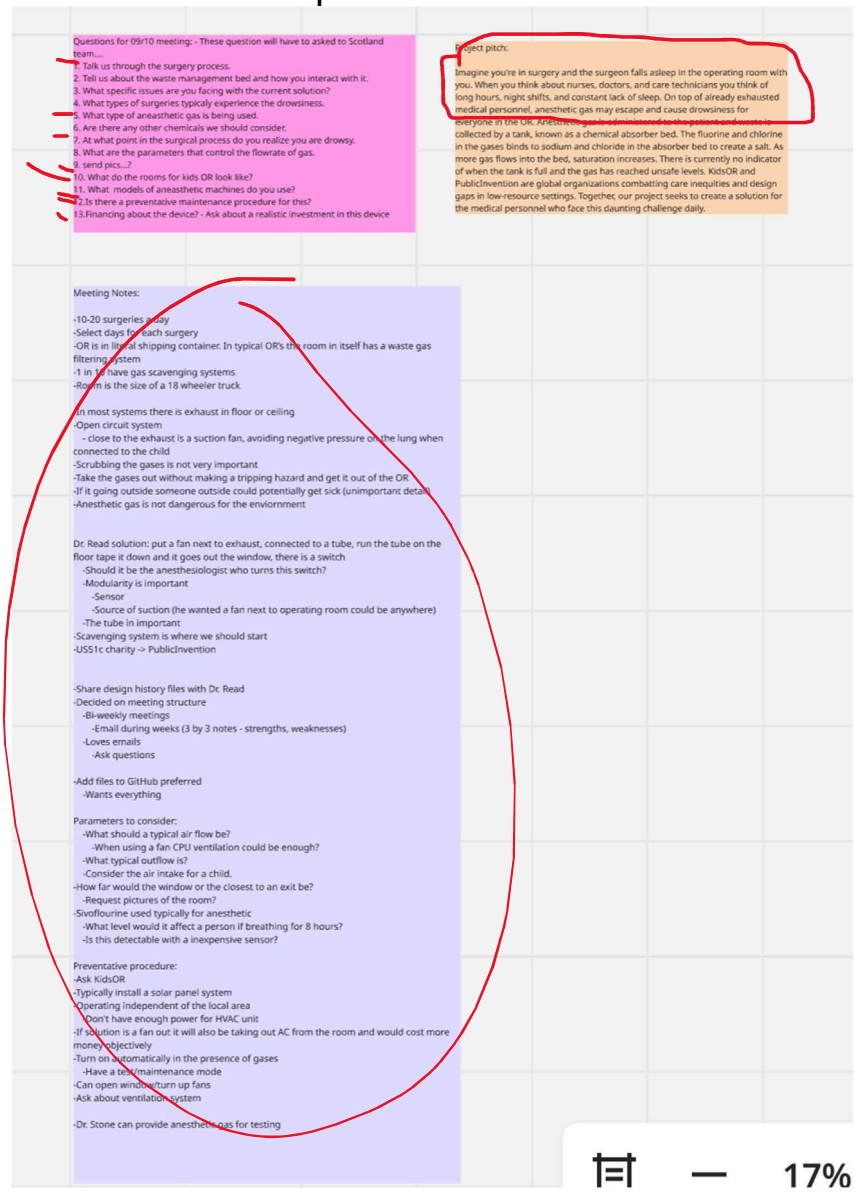


Figure 1: Miro Board with Project Pitch and Meeting Notes with Individual Contributions Circled

We created a project pitch for class that is on our Miro board. This will be helpful later in the DH01 document describing the purpose of the project and creating later project pitches for presentations. I also added a couple questions to ask the KidsOR team when we meet them so we can understand their current solution and how our eventual solution can fit into what they are doing. There is also a list of the meeting notes from the initial team meeting. These notes were transferred onto a word document and combined with the rest of the group, so we are all clear on expectations and what our stakeholder initially told us. All these additions cover the initial meet the stakeholder portion of the design process.

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Without keeping track of notes and comments from our stakeholder, we cannot start to create user needs.

Theoretical or Numerical Analysis Homework

4. Explain the range of quantitative input values that you would use and the range of values you would expect for the output. Describe how you determined these estimates.

   A potential anesthetic gas sensor that could be used for this project is the MPS4SENS Anesthesia Sensor made by Micro-Hybrid [1]. I found this by looking for anesthetic gas sensor datasheets, so I would be able to find an example voltage input value. This device specifically measures CO2, N2O, desflurane, and isoflurane. In the specifications, the listed DC output voltage is around 9.35mV. This value can be used as an example quantitative input value for voltage. Since this is a small voltage, a non-inverted amplifier could be used for the Arduino to read the signals. Assuming the Arduino input range is between 0V and 5V, I'm using 2.5V as a voltage output target. In this case, you would calculate A = Vout/ Vin = 2.5 / 0.00935 = 267. To calculate the resistor values for a non-inverting amplifier, A = 1 +R1/R2. The gain needs to be R1/R2 = 266. Based on this information, we could use a 266KOhm resistor as R1 and a 1 KOhm resistor as R2.

5. Bonus (2 pts): Solve the equations using the method or program of your choice and provide an example quantitative result to see if your intuition was correct from the question above.

   I plugged in the values below:

$R_1 = 266k\Omega$
$R_2 = 1k\Omega$
$V_{in} = 9.35mV$

$A = 1 + \frac{R_1}{R_2}$
$= 1 + \frac{266k\Omega}{1k\Omega}$
$= 1 + 266$
$= 267$ gain

$V_{out} = V_{in} A$
$0.00935(267)$
$= 2.50V$

Figure 2: Attachment from numerical analysis homework to highlight the equation used and the differences between sensors

When talking about this homework with my group mates we were able to start an interesting conversation about whether we would want to use non-dispersive infrared (NDIR) versus interference filter infrared (IFR) sensors. The solution I found above in this homework was an anesthetic waste gas sensor that was NDIR, while Kaayla based her calculations on a NIR sensor she found online. After doing further research I found that IFR sensors are a subset of NDIR sensors. NDIR sensors are the standard approach to anesthetic waste gas monitoring where a broad-spectrum of infrared light passes through a gas chamber and optical filters sort through these wavelengths before reaching the photodetector [10]. There are also Fourier-Transform Infrared (FTIR) sensors that could be a better solution, and they can capture all different gases at the same time while taking multiple wavelength measurements [10]. The NDIR sensors require multiple measurements for each gas and only takes in

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separate wavelength bands. The FTIR sensor is also said to calculate faster measurements and is portable, which matches our user needs [10]. This information will be helpful when we are choosing a sensor to build or looking for a sensor to use for our solution. During our stakeholder meeting it was also brought up that Dr. Bob has a sensor in mind. We have a meeting on Monday morning, so I will bring this information to him and see what his thoughts are.

Design Inputs Map

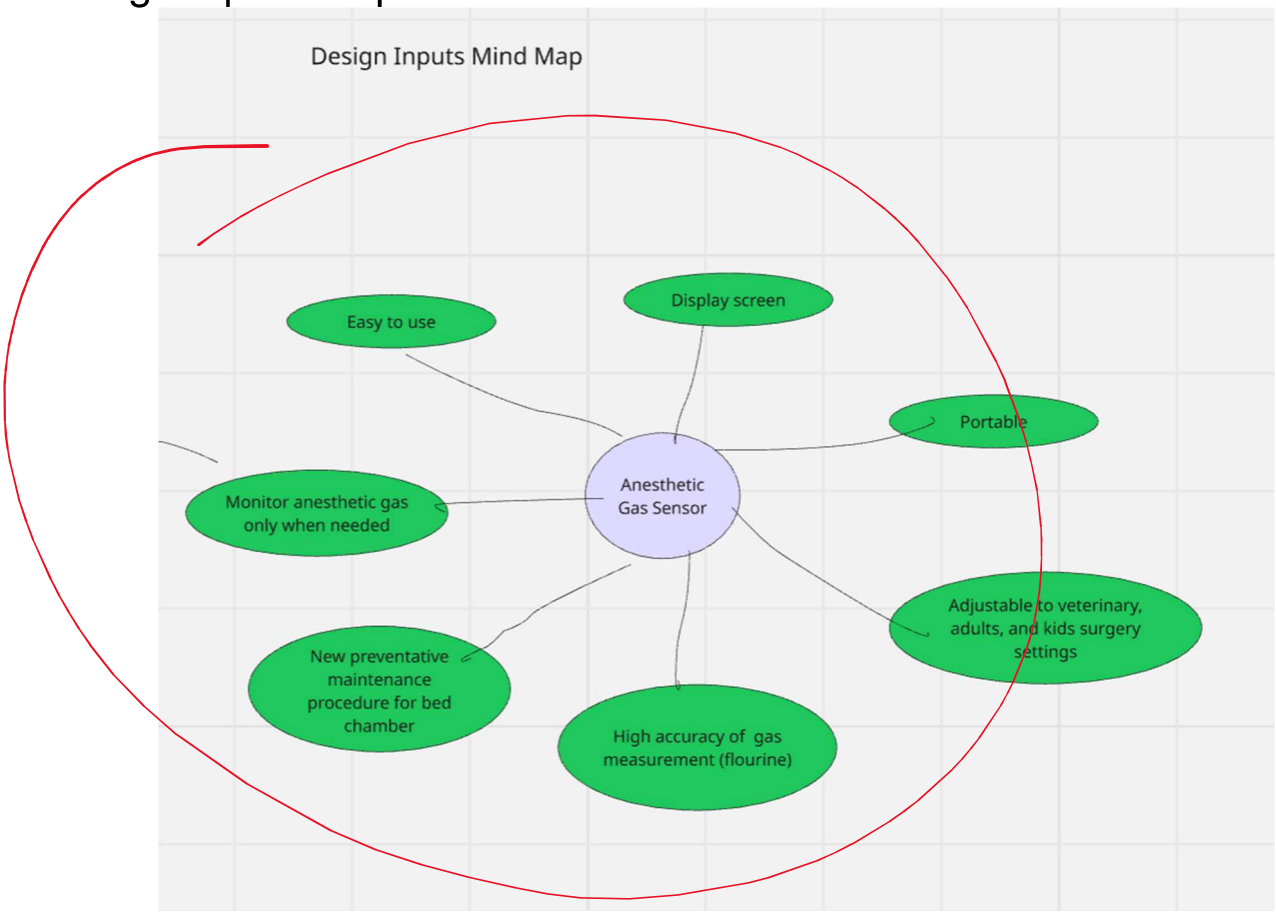


Figure 3: Design Inputs Map with Individual Contributions Circled

I started a design inputs mind map on our group's Miro board, so we can start getting a better picture of the project. Everything in green is a user need, our next step would be to add design inputs attached to the need, so we can further our understanding of the project. Once all members add to the map and some ideas are cut down, it will serve to be useful when brainstorming solutions and coming up with questions for the KidsOR team about their current solution. It will also help the group recenter if we get lost along the way in the design process.

Job Map Update

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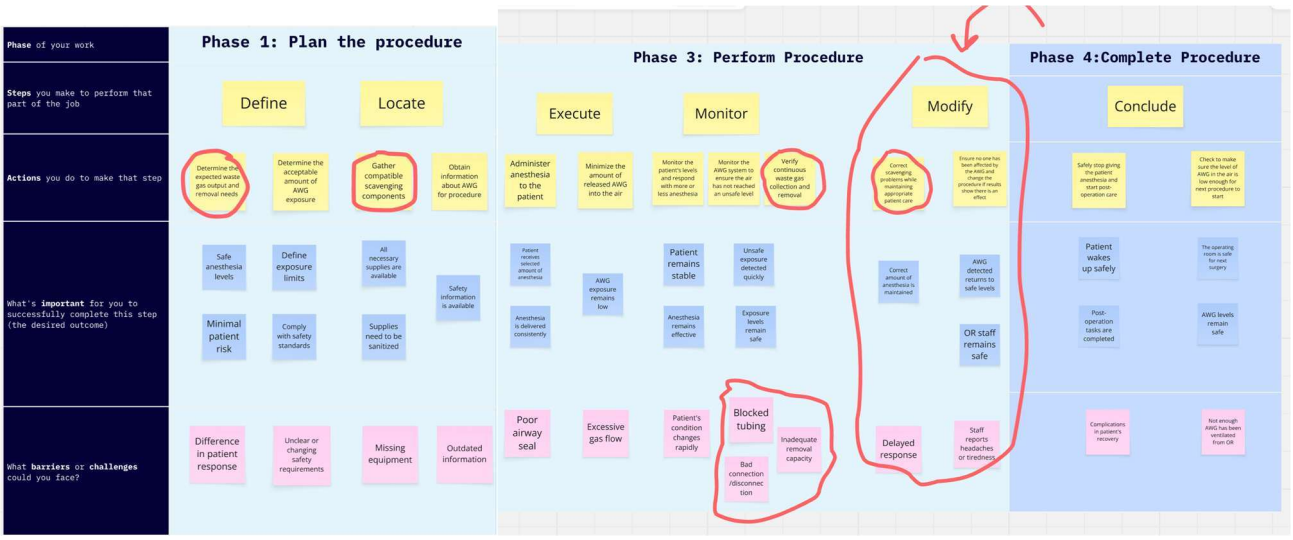
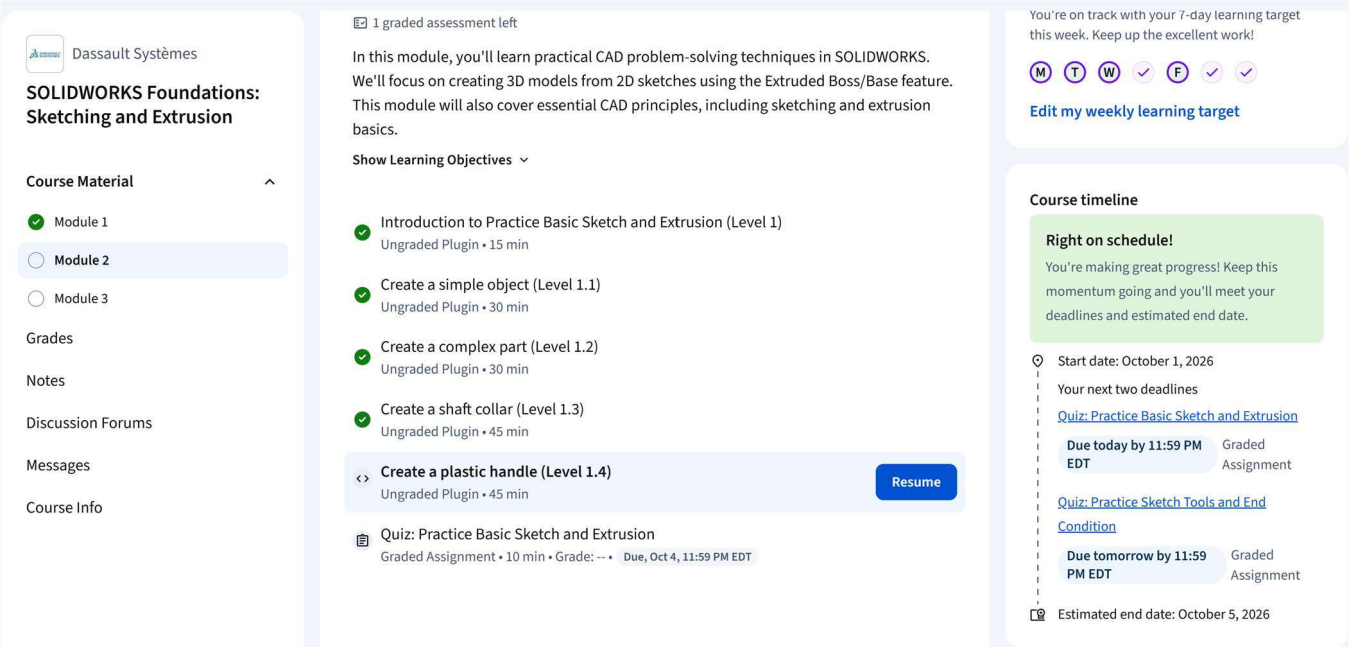


Figure 4: Updated Job Map Since DH01 with Changed Sections Circled

Since the meeting with KidsOR, the project we initially had in mind pivoted. I created an initial job map for DH01 with a sensor system in mind. Our new solution involves creating a standalone active scavenging system. Since our solutions were changed to have a passive design versus a more active design that could directly change the amount of AWG in the air, I also changed the job map a bit to make it align with the new ideas. Phase 2 stayed the same, so it is not pictured above. I also moved "modify" to phase 3 as part of performing a procedure because these adjustments can be made during the procedure not just afterwards. I changed a few challenges under the monitoring section to be specifically catered towards active scavenging systems. By understanding how an anesthesiologist works with the AWG system and how the system works during all phases of a procedure, we can create a solution that fits into the typical schedule of an anesthesiologist and make sure it changes this schedule minimally.

SolidWorks Tutorials



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Figure 5: Coursera Dessault Systemes SOLIDWORKS 3D CAD for Education Specialization

After meeting with KidsOR and getting new deliverables for a new solution, I realized that I needed to start working on improving my SolidWorks skills. I found Coursera when trying to find a cheaper way to get a six-sigma yellow belt certification/ specialization, so I could go to engineering expo and have a step up when finding manufacturing internships. I couldn't take the actual SolidWorks class because my organic chemistry lab interferes, but this website acts like a class. It has deliverables to complete every day, quizzes, homework, and discussion posts. By working on my foundational skills, I will be able to create a mockup model for the scavenging system, so we can use it to show stakeholders and our advisors what we have completed. As of right now I am 2 hours into this course and plan to complete this tutorial course by next engineering notebook. The courses include Sketching and Extrusion, Sketching and Reference Geometry, Revolution and Cut Features, Patterns and Sweeps, and Assemblies and Exam Level Part Modeling. In total this full course is 44 hours according to Coursera and sets you up to pass the Certified SolidWorks Associate (CSWA) Exam. I will take this exam realistically in late November or early December.

Anesthetic Waste Gas Equations

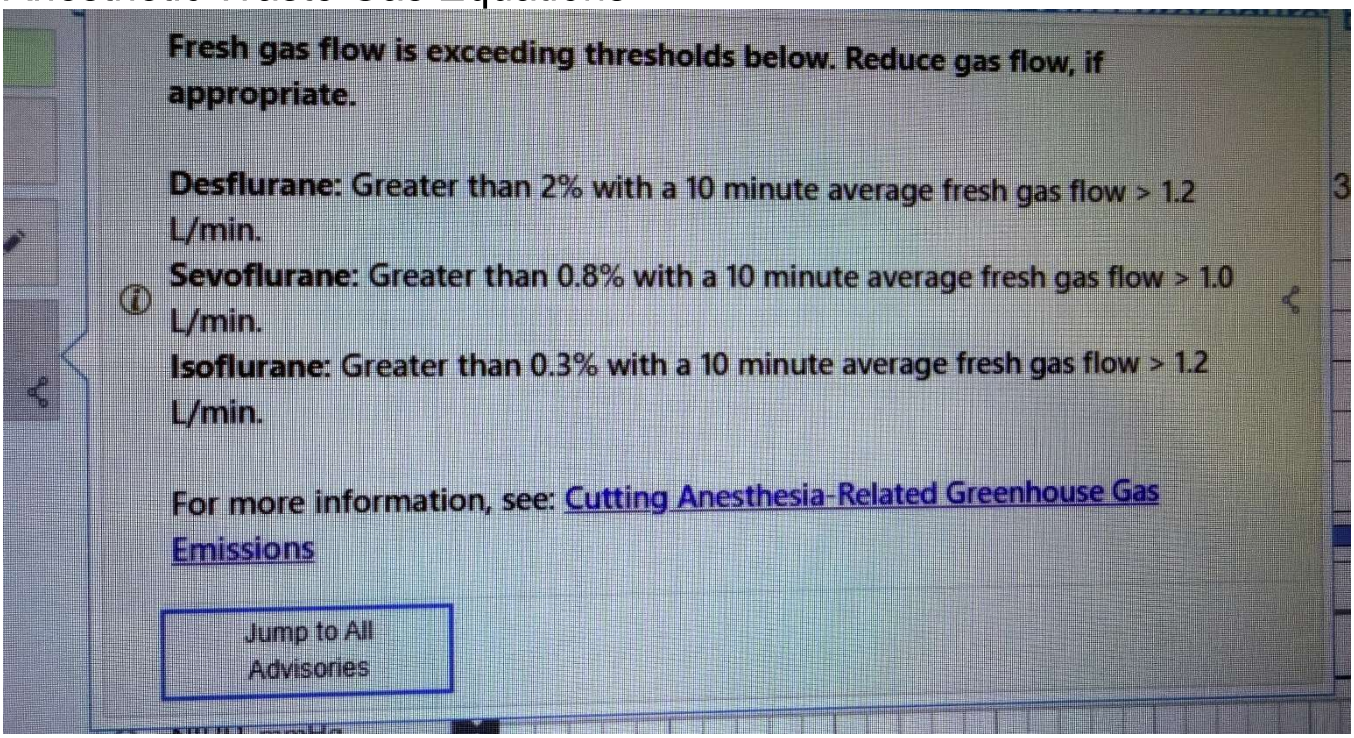


Figure 6: CHOP anesthetic gas low-flow versus high-flow alarm threshold system

The picture attached is directly from my interviewee that works at a hospital. I had asked him if he knew the thresholds for low versus high flow rates when he mentioned that alarm display will show up if there is high flow for a specific period of time. This information can give us an estimate of how much anesthetic gas is typically supplied during surgery. We can use the equation Gas flow x time = total gas. If the surgery is 1L/min for 60 minutes, like literature review 5, there is 60 liters of gas. When using sevoflurane one can multiple 0.8/100 x 1L x 60min = 0.48 L of sevoflurane vapor. The next step would be to figure out how much of this vapor turns into AWG to then also use this information to help understand how much gas needs to be collected by the scavenging system. This would also help figure

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out what type of preventive maintenance procedure we want to create after a specific amount of time the operating room is in use.

Low Fidelity Drawing 1

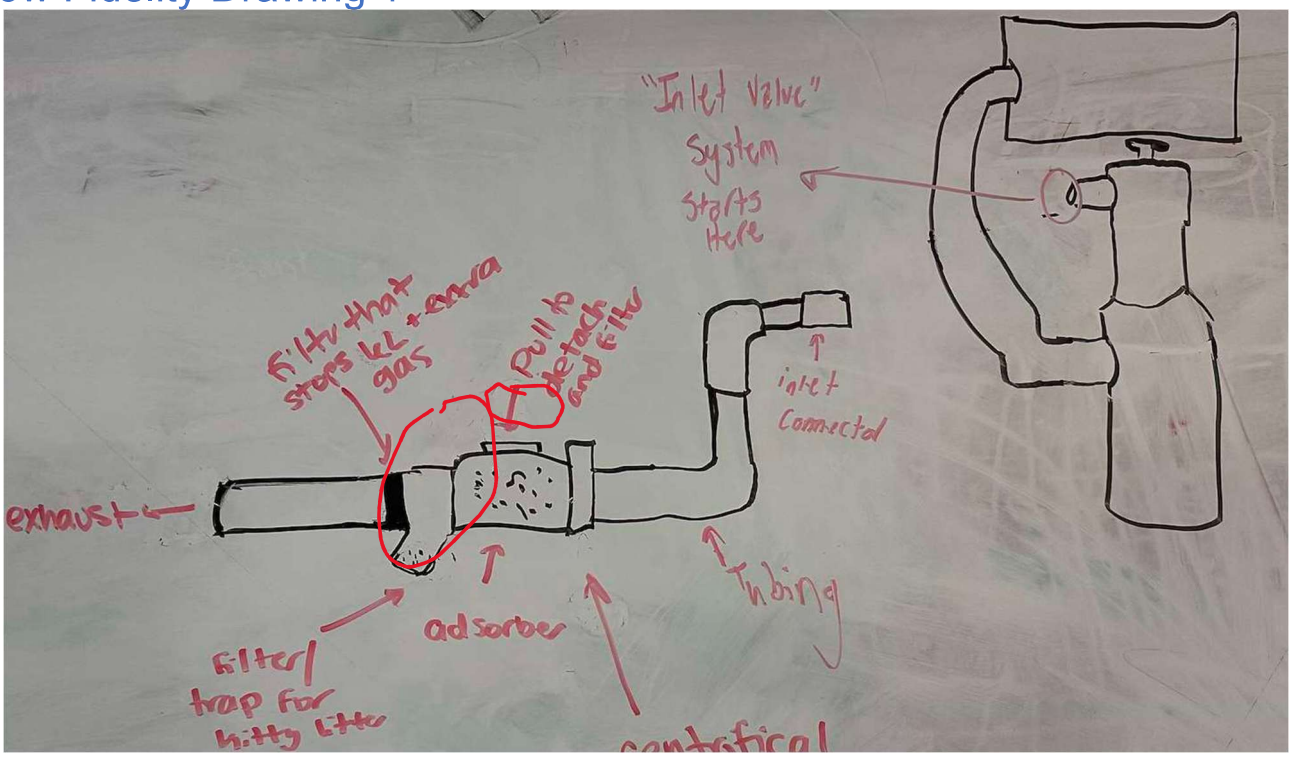


Figure 7: Low fidelity drawing we presented to stakeholders

During class time and our weekly team meeting, the team decided to draw out one singular idea to present to our stakeholders for review. This design allowed the stakeholders and the internal team to discuss future directions, if we were all on the same page about a solution, and if we were missing a certain element that may be needed. The first part of this design is the inlet valve. During one of our team meetings we found out where the standalone scavenging system should be connected to based on the manuals. In our meeting with the stakeholders later in the week, we were able to get picture confirmation of where the tubes they were using in the past were connected to. We also got clarification that the tubing is fully taken off the anesthetic machines because it's too much of an inconvenience for medical personnel to be walking around. This means that the anesthetic waste gas is going directly into the air and there is zero filtering or dilution of the gas, making even more sense why operating room teams are feeling tired. After the inlet valve, we plan to use PVC and two elbow pipes to make sure the new tubing system doesn't interfere with the negative pressure system already on the anesthetic machine that is connected to the inlet valve. The gas will then travel to the centrifugal fan. This type of fan was recommended to us by Dr. Stone to use. It works by drawing air into the system. We plan to use this fan to prevent gases from going back into the child's lungs and harming them. We also plan to make sure to reach the asked for 4L/min gas flow with the fan. After talking to the stakeholder team, we found out that the current system doesn't allow the gas to move down the tubes, so the gas just sits in the system. The fan will also make sure the gas can flow through the kitty litter adsorption system and the extra filter. After the fan is the kitty litter adsorption system. The kitty litter will adsorb the anesthetic gases. Once adsorbed, maintenance would be able to use the handle and lay the tray out in the sun. This will act as a cleaning system to reuse the kitty litter and reduce the long-term price. The next step we have in the system is a strainer. This is what I personally added to the system. Over the summer I

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worked with steam traps for a long time and I learned about a thermodynamic steam trap. Steam traps are used to trap purified steam and release condensate. Part of some thermodynamic steam traps is the addition of a strainer. This strainer works to collect any particles that may have been caught in the steam system. Typically, these do not need to be cleaned often and are a great addition. My idea behind this is to have a way to catch the kitty litter in case it moves past its filtering area. This way kitty litter buildup in the pipes isn't a concern and is easily fixable. With this in mind, we also have an extra filter that we were deciding on cloth or rubber to use. This would once again prevent the kitty litter from flowing down the pipes and causing buildup. It would also make sure the kitty litter is all being collected into the strainer/trap. At the very end we are showing the exhaust leaving the system. Drawing this out was super helpful. Our stakeholder said he was originally confused by our idea, but it now made sense. We were also able to get a lot of feedback on what we could change or make better. This initial design has helped the team centralize around one idea and it has aligned us for our next steps. During the design process we also decided it would be best to use PVC instead of ceramic, which was discussed during our advisor meeting. PVC comes in all different lengths and diameters. It is also used in construction already in low-resource areas. By choosing PVC, we are continuing to make our solution affordable and assessable for people in low-resource areas. Kitty litter is also being used because it is cheap, easy to find, and easy to use. Another potential option would be to use activated charcoal which works better with anesthetic gas; however, that is more expensive and harder to get. Both are viable options, but we ended up choosing kitty litter as our final initial solution. This idea also came from our meeting with Dr. Stone.

Low Fidelity Drawing 2

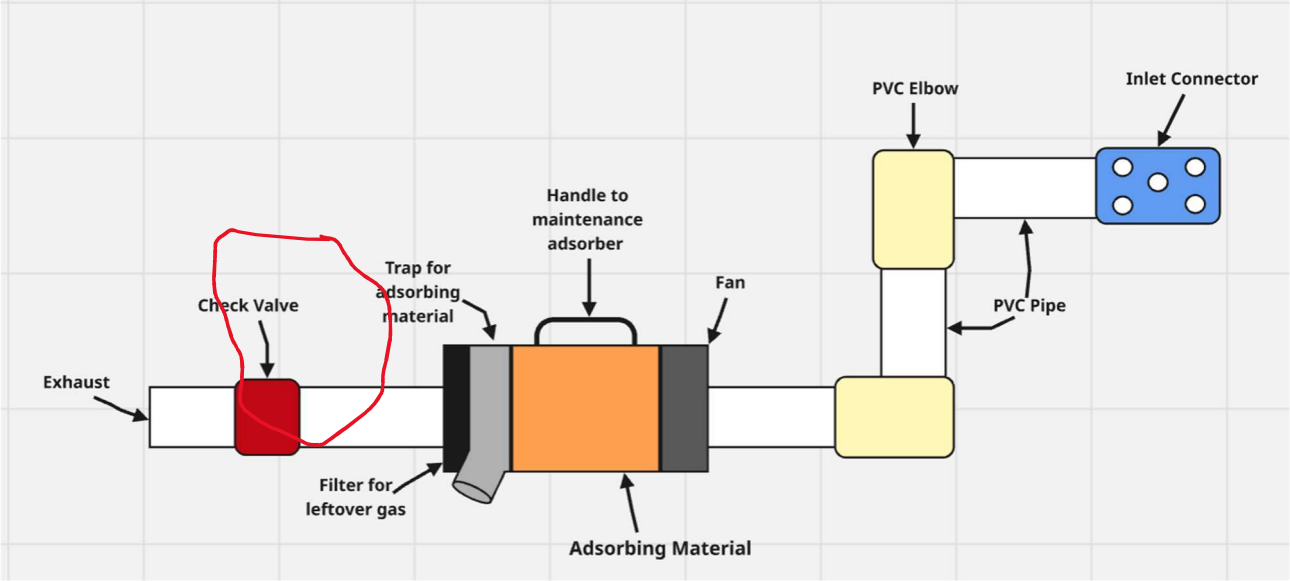


Figure 8: Low fidelity prototype drawing for PDR

Kaayla made this drawing to further clean up our initial drawing. It was also created after our meeting with our stakeholders and presented our idea on the first iteration of the solution. During that meeting, the stakeholders brought up how we may need to dilute the gas prior to it even reaching the adsorber. This would allow the gas to be more easily adsorbed once it passes through the kitty litter. Another question that was brought up was brought up during the meeting is whether kitty litter works for all types of gases. One of the gases it may not work with is sevoflurane, which is one of the most used gases by anesthesiologists during surgery. Kitty litter does work for desflurane and isoflurane which are also common anesthetic gases. The question the group needs to find out now is if there is something that works better and is just as cheap as kitty litter. This brings back into question the activated

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charcoal and if the extra price tag is worth adsorbing another gas for. This also brings up the idea of the holes in the tube to dilute the gases. If sevoflurane is specifically not being adsorbed, but it is being diluted one way or another, could our original solution still work? Once again, this will be discussed during our team meeting on Wednesday or Tuesday to confirm if we are using it. Dr. Andy also brought up the idea of a check valve to help prevent against backflow. Backflow means that there is water moving back through the piping and potentially into a child's lungs in this case. The water is made because of moisture from the child's breathing. By adding a check valve, this would ensure that backflow isn't occurring. My addition to this was confirming from the engineering notebook article 7 that by adding a check valve, backflow would be prevented, and moisture wouldn't cause issues for the child patient. My comment led to the official addition of a check valve in the PDR solution. Overall, this model is a cleaner version with new additions from our stakeholder feedback. In the end we now have a system that is safer and has internal checks for backflow and additional gas dilution.

Low Fidelity Prototype 1

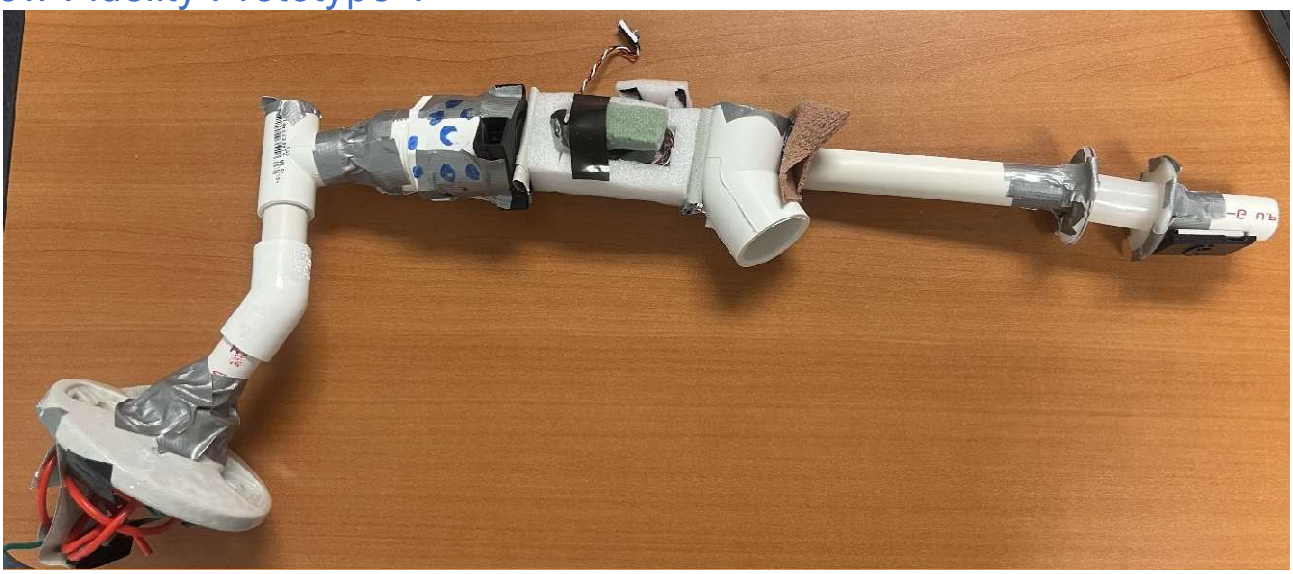


Figure 9: Low fidelity prototype created for PDR

Rachel and I created this using what we found in the lab and boxes that once held ordered items. The entire prototype is replicating the PDR drawing into something real. The left circle piece is the inlet valve. For this we used a lightbulb holder and taped it to a piece of short PVC piping. We then used two elbow pieces to represent system we have created being moved away from the anesthetic machine, so it isn't in the way of anything already there. The piece with blue dots represents the holes before the fan to make sure the gas is being diluted. We also want it close to the fan so anesthetic gas isn't just going out through the holes. The black piece that is next to the blue dots and is slightly covered in tape is the fan. Dr. Stone gave us this fan that we could use for modeling the week prior. The next is the Styrofoam box that holds the green adsorbent and a pH monitoring system which is being represented through the wires leaving the box. One can also see at the top of the box, connected in black tape, is the handle for the filter system that can take the box in and out. The next part is the strainer and after that is a piece of athletic tape, we used to represent the rubber or cotton additional prevention. At the end of the pipe, one can see that there is an addition between the two PVC pipes. This addition is meant to represent what a check valve looks like. By creating a low-fidelity prototype we are creating something that is real and we can physically move and touch. This can start to get us into the future building phase as well as seeing how far our final solution would come from this initial put together. By creating this and showing

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our stakeholders, we are able to show that we are moving at a good pace with this project and are ready for more feedback.

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